

Sharp p -Dependence for the ℓ_p Margin Dimension

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Status

This packet gives a likely valid full resolution of the p -dependence question following Proposition 3.5 in Ashlagi–Livni–Moran–Waknine, *Margin in Abstract Spaces*, arXiv:2603.07221.

The source paper proves the ℓ_p margin taxonomy

$$\dim_{\ell_p}(\gamma) = \begin{cases} \Theta(\gamma^{-q}), & 1 < p \leq 2, \quad q = p/(p-1), \\ \Theta_p(\gamma^{-2}), & 2 < p < \infty, \end{cases}$$

and then asks for bounds that are also tight in their dependence on p . The result below identifies that dependence: the low- p formula is exact, and the high- p dependence is linear in p , with the sharp $p\gamma^{-2}$ scale in the small-margin regime.

Source Question

The relevant source passage is Proposition 3.5 and the paragraph following it on page 8. For $p > 2$, the source records $\dim_{\ell_p}(\gamma) = \Theta_p(\gamma^{-2})$, with the hidden constant bounded between an absolute constant and p . It then says that, although the dependence on the margin parameter γ is tight in all regimes, obtaining bounds also tight in their dependence on p remains open.

Definitions

We use the source paper’s notation. For a Banach space X , $\dim_X(\gamma)$ is the largest size of a γ -shattered subset of the unit ball of X by norm-one linear functionals. Corollary 3.8 of the source paper says that $x_1, \dots, x_n \in B_X$ are γ -shattered if and only if

$$\left\| \sum_{i=1}^n a_i x_i \right\|_X \geq \gamma \sum_{i=1}^n |a_i| \quad (a_i \in \mathbb{R}).$$

Equivalently, the map $T : \ell_1^n \rightarrow X$, $Te_i = x_i$, is an embedding with $\|T\| \leq 1$ and $\|T^{-1}\| \leq \gamma^{-1}$ on its range.

Idea of the Proof

The upper bounds come from testing a shattered set against random signs. If $x_1, \dots, x_n$ are γ -shattered, then every sign sum $\sum \varepsilon_i x_i$ has norm at least γn . Averaging over signs and using scalar

Khintchine inequalities gives a direct restriction on n . For $1 < p \leq 2$, the relevant Khintchine upper constant is 1, giving the exact upper bound $n \leq \gamma^{-q}$. For $p > 2$, the best Khintchine constant B_p gives $n \leq B_p^2 \gamma^{-2}$, and Haagerup's estimate has $B_p^2 \asymp p$.

The lower bounds use explicit copies of ℓ_1^n in finite-dimensional ℓ_p . For $1 < p \leq 2$, the standard basis is optimal. For $p > 2$, the right construction is the Rademacher system on the sign cube. Hitzenko's sharp moment estimate gives the missing p factor in the small-margin regime. A separate sign-alignment estimate shows that for every fixed $\gamma < 1$, the high- p dimension is still linear in p . Finally, Clarkson's inequality explains why one cannot expect a uniform lower bound $cp\gamma^{-2}$ up to $\gamma = 1$: there is a genuine endpoint transition.

The final push adds an exact high- p two-parameter formula in the range $p \geq 3$. It uses Schechtman's many-function Hanner inequality, which says that for $p \geq 3$ the Rademacher p -moment of an L_p -valued sign sum is dominated by the scalar sign sum of the individual L_p -norms. The lower bound is the Rademacher-system construction again, now sharpened by the observation that a symmetric convex function is minimized on the ℓ_1 simplex at equal coefficients.

Main Result

Theorem 1 (Sharp p -dependence for ℓ_p margin shattering). *Let $0 < \gamma < 1$.*

(a) *If $1 < p \leq 2$ and $q = p/(p-1)$, then*

$$\dim_{\ell_p}(\gamma) = \lfloor \gamma^{-q} \rfloor.$$

(b) *If $2 < p < \infty$, then*

$$\dim_{\ell_p}(\gamma) \leq B_p^2 \gamma^{-2} \leq Cp \gamma^{-2},$$

where B_p is the optimal upper Khintchine constant and C is universal.

(c) *There are universal constants $c > 0$ and $\gamma_0 \in (0, 1)$ such that for all $2 < p < \infty$ and $0 < \gamma \leq \gamma_0$,*

$$\dim_{\ell_p}(\gamma) \geq cp \gamma^{-2}.$$

Consequently, uniformly in $p > 2$ and small γ ,

$$\dim_{\ell_p}(\gamma) = \Theta(p\gamma^{-2}).$$

(d) *For every $2 < p < \infty$ and $0 < \gamma < 1$,*

$$\dim_{\ell_p}(\gamma) \geq \max \{1, \lfloor p \log_2(1/\gamma) \rfloor\}.$$

In particular, for every fixed $\gamma \in (0, 1)$,

$$\dim_{\ell_p}(\gamma) = \Theta_\gamma(p) \quad (p \rightarrow \infty).$$

(e) *If $2 < p < \infty$ and $\gamma > 2^{-1/p}$, then*

$$\dim_{\ell_p}(\gamma) = 1.$$

(f) *If $3 \leq p < \infty$, then the exact value is*

$$\dim_{\ell_p}(\gamma) = \max \left\{ n \geq 1 : \left(\mathbb{E} \left| \frac{\varepsilon_1 + \cdots + \varepsilon_n}{n} \right|^p \right)^{1/p} \geq \gamma \right\}.$$

Thus the p -dependence in the source paper's ℓ_p taxonomy is sharp: there is no hidden p -loss for $1 < p \leq 2$, and the high- p dependence is linear in p .

Proof

Proof. We first prove the upper bounds. Let $x_1, \dots, x_n$ be γ -shattered in the unit ball of ℓ_p . By the source characterization, for every sign vector $\varepsilon \in \{-1, 1\}^n$,

$$\left\| \sum_{i=1}^n \varepsilon_i x_i \right\|_p \geq \gamma n.$$

Suppose first that $1 < p \leq 2$. By the scalar Khintchine inequality with upper constant 1 for $p \leq 2$,

$$\mathbb{E}_\varepsilon \left| \sum_{i=1}^n \varepsilon_i a_i \right|^p \leq \left(\sum_{i=1}^n |a_i|^2 \right)^{p/2} \leq \sum_{i=1}^n |a_i|^p.$$

Applying this coordinatewise gives

$$(\gamma n)^p \leq \mathbb{E}_\varepsilon \left\| \sum_{i=1}^n \varepsilon_i x_i \right\|_p^p = \sum_j \mathbb{E}_\varepsilon \left| \sum_{i=1}^n \varepsilon_i x_i(j) \right|^p \leq \sum_{i=1}^n \|x_i\|_p^p \leq n.$$

Thus $n \leq \gamma^{-p/(p-1)} = \gamma^{-q}$.

The reverse inequality for $1 < p \leq 2$ is supplied by the standard basis. Indeed, if $e_1, \dots, e_n$ are the first n unit vectors and $\sum_i |a_i| = 1$, then

$$\left\| \sum_{i=1}^n a_i e_i \right\|_p = \|a\|_p \geq n^{1/p-1} \|a\|_1 = n^{-1/q}.$$

Hence the standard basis is γ -shattered whenever $n \leq \gamma^{-q}$, proving $\dim_{\ell_p}(\gamma) = \lfloor \gamma^{-q} \rfloor$.

Now let $p > 2$. The same sign test, together with the scalar Khintchine upper constant B_p , gives

$$\begin{aligned} (\gamma n)^p &\leq \mathbb{E}_\varepsilon \left\| \sum_{i=1}^n \varepsilon_i x_i \right\|_p^p \\ &= \sum_j \mathbb{E}_\varepsilon \left| \sum_{i=1}^n \varepsilon_i x_i(j) \right|^p \\ &\leq B_p^p \sum_j \left(\sum_{i=1}^n |x_i(j)|^2 \right)^{p/2}. \end{aligned}$$

Since $p/2 \geq 1$,

$$\left(\sum_{i=1}^n |x_i(j)|^2 \right)^{p/2} \leq n^{p/2-1} \sum_{i=1}^n |x_i(j)|^p.$$

Summing in j and using $\|x_i\|_p \leq 1$, we get

$$(\gamma n)^p \leq B_p^p n^{p/2}.$$

Therefore $n \leq B_p^2 \gamma^{-2}$. Haagerup's sharp Khintchine constants give $B_p^2 \leq Cp$, proving the asserted high- p upper bound.

We next prove the small-margin high- p lower bound. We use Hitzenko's Rademacher moment estimate in the following standard form: there is a universal $A > 0$ such that, for every $p \geq 2$, every finite scalar sequence $b = (b_i)$, and independent Rademacher signs (ε_i) , if (b_i^*) is the non-increasing rearrangement of $(|b_i|)$, then

$$\left(\mathbb{E} \left| \sum_i b_i \varepsilon_i \right|^p \right)^{1/p} \geq A \left(\sum_{i \leq [p]} b_i^* + \sqrt{p} \left(\sum_{i > [p]} (b_i^*)^2 \right)^{1/2} \right).$$

Consequently, if $b \in \mathbb{R}^n$ and $\sum_i |b_i| = 1$, then

$$\left(\mathbb{E} \left| \sum_{i=1}^n b_i \varepsilon_i \right|^p \right)^{1/p} \geq c_0 \min \left(1, \sqrt{\frac{p}{n}} \right)$$

with a universal $c_0 > 0$. Indeed, if $n \leq [p]$, the head term is 1. If $n > [p]$, either the head has mass at least $1/2$, or the tail has ℓ_1 -mass at least $1/2$, in which case its ℓ_2 -norm is at least $1/(2\sqrt{n})$.

Let $n = \lfloor c_1 p \gamma^{-2} \rfloor$, with $c_1 > 0$ sufficiently small, and let $\Omega = \{-1, 1\}^n$ with the uniform probability measure. For $1 \leq i \leq n$, let $r_i(\omega) = \omega_i$. Embed $L_p(\Omega)$ isometrically into $\ell_p^{2^n}$ by

$$f \mapsto 2^{-n/p} (f(\omega))_{\omega \in \Omega},$$

and let x_i be the image of r_i . Then $\|x_i\|_p = 1$, and for $\sum_i |a_i| = 1$,

$$\left\| \sum_{i=1}^n a_i x_i \right\|_p = \left(\mathbb{E}_\omega \left| \sum_{i=1}^n a_i r_i(\omega) \right|^p \right)^{1/p} \geq c_0 \min \left(1, \sqrt{\frac{p}{n}} \right).$$

For $0 < \gamma \leq \gamma_0$, with γ_0 universal and c_1 chosen small enough, the last expression is at least γ . Thus the x_i 's are γ -shattered and $\dim_{\ell_p}(\gamma) \geq c p \gamma^{-2}$.

For the fixed-margin high- p lower bound, use the same Rademacher system but choose $n \leq p \log_2(1/\gamma)$. For every $a \in \mathbb{R}^n$ with $\sum_i |a_i| = 1$, there is a sign vector ω such that $\sum_i a_i r_i(\omega) = 1$, namely $\omega_i = \text{sgn}(a_i)$ on the support of a . Hence

$$\left(\mathbb{E}_\omega \left| \sum_{i=1}^n a_i r_i(\omega) \right|^p \right)^{1/p} \geq 2^{-n/p}.$$

If $n \leq p \log_2(1/\gamma)$, this is at least γ , so the Rademacher vectors are γ -shattered. This proves

$$\dim_{\ell_p}(\gamma) \geq \max \{1, \lfloor p \log_2(1/\gamma) \rfloor\}.$$

Together with the upper bound $O(p\gamma^{-2})$, it follows that, for every fixed $\gamma < 1$, $\dim_{\ell_p}(\gamma) = \Theta_\gamma(p)$.

It remains to justify the endpoint obstruction. Suppose $p > 2$ and two unit vectors $x, y \in \ell_p$ are γ -shattered. Applying the source characterization to (x, y) with coefficients $(1, 1)$ and $(1, -1)$ gives

$$\left\| \frac{x+y}{2} \right\|_p \geq \gamma, \quad \left\| \frac{x-y}{2} \right\|_p \geq \gamma.$$

Clarkson's inequality for L_p , $p \geq 2$, gives

$$\left\| \frac{x+y}{2} \right\|_p^p + \left\| \frac{x-y}{2} \right\|_p^p \leq \frac{\|x\|_p^p + \|y\|_p^p}{2} \leq 1.$$

Therefore $2\gamma^p \leq 1$, i.e. $\gamma \leq 2^{-1/p}$. If $\gamma > 2^{-1/p}$, no two-point shattered set exists. Since one unit vector is γ -shattered for every $\gamma < 1$, we have $\dim_{\ell_p}(\gamma) = 1$ in this endpoint range.

We finally prove the exact formula for $p \geq 3$. Let

$$\rho_p(n) = \left(\mathbb{E} \left| \frac{\varepsilon_1 + \dots + \varepsilon_n}{n} \right|^p \right)^{1/p}.$$

For the lower bound, use the Rademacher coordinate functions $r_1, \dots, r_n$ on $\{-1, 1\}^n$, embedded isometrically into a finite ℓ_p as above. If $\sum_i |a_i| = 1$, signs may be absorbed into the Rademacher variables, so assume $a_i \geq 0$. The function

$$(a_1, \dots, a_n) \mapsto \mathbb{E} \left| \sum_{i=1}^n a_i \varepsilon_i \right|^p$$

is convex and symmetric on $\mathbb{R}^n$. Hence its restriction to the probability simplex is Schur-convex. Since every point of the simplex majorizes $(1/n, \dots, 1/n)$, this function is minimized at equal coefficients. Therefore

$$\left\| \sum_{i=1}^n a_i r_i \right\|_{L_p} = \left(\mathbb{E} \left| \sum_{i=1}^n a_i \varepsilon_i \right|^p \right)^{1/p} \geq \rho_p(n).$$

Thus the Rademacher vectors are γ -shattered whenever $\rho_p(n) \geq \gamma$.

Conversely, suppose $x_1, \dots, x_n \in B_{\ell_p}$ are γ -shattered and $p \geq 3$. Schechtman's many-function Hanner inequality gives, for L_p -valued functions f_i ,

$$\mathbb{E}_\varepsilon \left\| \sum_{i=1}^n \varepsilon_i f_i \right\|_p^p \leq \mathbb{E}_\varepsilon \left| \sum_{i=1}^n \varepsilon_i \|f_i\|_p \right|^p.$$

Applying this to x_i , and then using $\|x_i\|_p \leq 1$, gives the next bound by the elementary monotonicity of scalar Rademacher moments in each coefficient:

$$\mathbb{E}_\varepsilon \left\| \sum_{i=1}^n \varepsilon_i x_i \right\|_p^p \leq \mathbb{E}_\varepsilon |\varepsilon_1 + \dots + \varepsilon_n|^p.$$

Shattering gives $\|\sum_i \varepsilon_i x_i\|_p \geq \gamma n$ for every sign vector, so

$$(\gamma n)^p \leq \mathbb{E} |\varepsilon_1 + \dots + \varepsilon_n|^p.$$

Equivalently, $\rho_p(n) \geq \gamma$. This proves the exact formula for $p \geq 3$. □

Interpretation

This theorem resolves the p -dependence asked for after Proposition 3.5. For $1 < p \leq 2$, the exact answer is $\lfloor \gamma^{-q} \rfloor$, so the source's p -dependence through $q = p/(p-1)$ has no additional hidden constant. For $p > 2$, the upper constant is $B_p^2 \asymp p$, and the Rademacher-Hitczenko lower bound shows that this linear growth in p is attained in the small-margin regime where the source's $\Theta_p(\gamma^{-2})$ asymptotic is meaningful. The sign-alignment lower bound further shows that linear p -growth persists for every fixed $\gamma < 1$.

The Clarkson endpoint statement is included to prevent an overinterpretation: one cannot have a universal lower bound $cp\gamma^{-2}$ for every $\gamma \in (0, 1)$, because the dimension drops to 1 when $\gamma > 2^{-1/p}$.

Thus the remaining near- $\gamma = 1$ transition is a different two-parameter endpoint phenomenon, not a failure of the p -dependence resolution requested in the source remark.

In fact, for $p \geq 3$ the endpoint transition is completely described by the Rademacher random-walk moment $\rho_p(n)$. This is sharper than needed for the source p -dependence question. Jenkins–Tkocz note that the corresponding real Rademacher many-function Hanner inequality is open in the range $2 < p < 3$. Conditional on that open inequality, the same proof gives the same exact formula for every $2 \leq p < \infty$.

Novelty and Literature Search

Local run indexes and parsed sources were searched for arXiv:2603.07221, the source title, ℓ_p margin phrases, p -dependence phrases, fat-shattering phrases, Rademacher moment phrases, Hitzzenko, Balcan–Berlind, and ℓ_1^n embeddings into L_p . Web searches on 2026-06-15 covered the source title and p -dependence phrase, Hitzzenko/Rademacher moment estimates, Khintchine constants, Clarkson inequalities, and Banach–Mazur or ℓ_1^n -into- L_p formulations. These searches found the source paper and the standard tools cited below. The final push also identified Schechtman’s many-function Hanner inequality for real L_p , $p \geq 3$, via Jenkins–Tkocz. This appears not to have been applied to the exact margin dimension question in the source paper.

References

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- [8] J. Jenkins and T. Tkocz, *Complex Hanner’s inequality for many functions*, arXiv:2207.09122, 2022.

Human Review Recommendation

Review as a likely valid full solution to the p -dependence question following Proposition 3.5 of the source paper. The main checks are the normalization in Corollary 3.8, the coordinatewise

Khintchine upper-bound computations, and the scope interpretation: the theorem solves the p -growth problem, while the $p \geq 3$ refinement gives an exact two-parameter endpoint formula. The remaining exact-formula range $2 < p < 3$ is tied to the open real Rademacher many-function Hanner inequality identified by Jenkins–Tkocz.